

A SHORT PROOF THAT LINK-IRREGULAR TOURNAMENTS EXIST FOR EVERY ORDER AT LEAST SIX

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ABSTRACT. A digraph is link-irregular if no two of its vertices have isomorphic links. Bastien and Khormali conjectured that a link-irregular tournament on n vertices exists if and only if $n \geq 6$, and Chau proved it by substitution into a transitive tournament. We give a short different proof, by a construction that adds two vertices at a time.

1. INTRODUCTION

The link $L_D(v)$ of a vertex v in a digraph D is the subdigraph induced by the in- and out-neighbours of v , and D is *link-irregular* if $L_D(u) \not\cong L_D(v)$ for all distinct vertices u, v . Links of vertices in graphs were introduced by Ali, Chartrand and Zhang [1], and Bastien and Khormali [2] carried the notion over to digraphs. In a tournament every other vertex is a neighbour of v , so the link of v is the subtournament $T - v$. A tournament is therefore link-irregular exactly when its n vertex-deleted subtournaments are pairwise non-isomorphic.

Bastien and Khormali showed that there is no link-irregular tournament on $n \leq 5$ vertices (for $n = 5$ there are only four tournaments on four vertices, up to isomorphism, to serve as links), gave link-irregular tournaments on 6, 7 and 8 vertices, and found examples by computer search for every $n \leq 100$. Their construction adds one dominating vertex at a time and stops working at 9 vertices. They made the following conjecture ([2, Conjecture 6]; Conjecture 3.7 in the journal version [3]).

Conjecture 1. A link-irregular tournament on n vertices exists if and only if $n \geq 6$.

Here $n \geq 2$; the tournament with one vertex is link-irregular for want of a second vertex.

Chau [4] proved Conjecture 1. Call a tournament 2-strong if it is strongly connected and stays so after any one vertex is deleted. Chau showed that substituting link-irregular 2-strong tournaments into a transitive tournament gives a link-irregular tournament, and found such tournaments of every order from 8 to 15 by computer search; together with the examples of Bastien and Khormali this covers every $n \geq 6$. Harte formalised Chau's proof in Lean [6] and showed that 7 is the least order of a link-irregular 2-strong tournament [5].

We give a second proof. Its only finite input is a pair of tournaments on six and seven vertices whose links can be told apart by hand.

Theorem 1.1. *For every $n \geq 6$ there is a link-irregular tournament on n vertices that has neither a source nor a sink.*

The proof is an induction from n to $n + 2$ (Lemma 2.1), started at $n = 6$ and $n = 7$ (Section 3). The new pair of vertices is a source and a sink of the old tournament joined by one reversed arc; this pair can be recognised inside every link, which is what makes the step work.

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We write $u \rightarrow v$ when the arc between u and v points from u to v , and $\text{sc}_T(v)$ for the score (out-degree) of v in a tournament T . A source of T is a vertex of score $|T| - 1$ and a sink is a vertex of score 0. Isomorphisms of tournaments preserve scores.

2. THE STEP FROM n TO $n + 2$

Let T be a tournament on a vertex set V with $|V| = n$. Let T^+ be the tournament on $V \cup \{s, t\}$ that contains T and has the arcs

$$s \rightarrow v \text{ and } v \rightarrow t \quad (v \in V), \quad t \rightarrow s$$

(Figure 1). In T^+ we have $\text{sc}(s) = n$, $\text{sc}(t) = 1$ and $\text{sc}(v) = \text{sc}_T(v) + 1$ for $v \in V$.

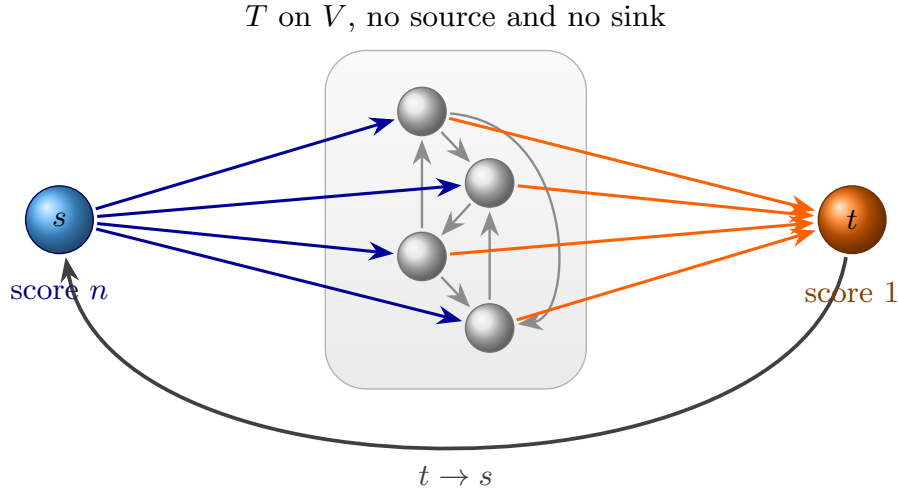


FIGURE 1. The tournament T^+ : s beats every vertex of T , every vertex of T beats t , and t beats s .

Lemma 2.1. *Let $n \geq 3$ and let T be a link-irregular tournament on n vertices with neither a source nor a sink. Then T^+ is a link-irregular tournament on $n + 2$ vertices with neither a source nor a sink.*

Proof. Since $1 \leq \text{sc}_T(v) \leq n - 2$ for $v \in V$, the scores in T^+ lie between 1 and n , while T^+ has $n + 2$ vertices; so T^+ has no source and no sink. We compare the links $T^+ - x$ for $x \in V \cup \{s, t\}$, each of which has $n + 1$ vertices, so that a source in it has score n .

The link of s . In $T^+ - s$ the vertex t has score 0, so $T^+ - s$ has a sink. Every $v \in V$ has score $\text{sc}_T(v) + 1 \leq n - 1$ there, so $T^+ - s$ has no source.

The link of t . In $T^+ - t$ the vertex s has score n , so $T^+ - t$ has a source. Every $v \in V$ has score $\text{sc}_T(v) \geq 1$ there, so $T^+ - t$ has no sink.

The link of $v \in V$. In $T^+ - v$ the vertex s has score $n - 1$, the vertex t has score 1, and $u \in V \setminus \{v\}$ has score $\text{sc}_{T-v}(u) + 1$, which lies between 1 and $n - 1$. So $T^+ - v$ has neither a source nor a sink.

Hence the link of s , the link of t and the links of vertices of V fall into three classes that no isomorphism can mix: the first has a sink but no source, the second a source but no sink, the third neither.

It remains to show $T^+ - v \not\cong T^+ - w$ for distinct $v, w \in V$. Call a vertex y of $T^+ - v$ *marked* if $\text{sc}(y) = 1$ and the unique out-neighbour of y has score $n - 1$. The vertex t is marked, its out-neighbour being s . Any other vertex of score 1 in $T^+ - v$ is some $u \in V \setminus \{v\}$ with $\text{sc}_{T-v}(u) = 0$; its only out-neighbour is t , whose score is $1 \neq n - 1$ because $n \geq 3$. So t is the only marked vertex of $T^+ - v$, and s is its out-neighbour. The same holds in $T^+ - w$. An isomorphism $\varphi: T^+ - v \rightarrow T^+ - w$ preserves scores and arcs, hence maps the marked vertex to the marked vertex, so $\varphi(t) = t$ and then $\varphi(s) = s$. Its restriction is an isomorphism from $T - v$ onto $T - w$, which is impossible because T is link-irregular. $\square$

3. TWO STARTING TOURNAMENTS

Let T_6 be the tournament on $\{1, \dots, 6\}$ with the arcs

$$1 \rightarrow 2, 3, 4, 5; \quad 2 \rightarrow 3, 4, 5, 6; \quad 3 \rightarrow 4, 6; \quad 4 \rightarrow 5, 6; \quad 5 \rightarrow 3; \quad 6 \rightarrow 1, 5,$$

where $i \rightarrow j, k$ means $i \rightarrow j$ and $i \rightarrow k$; the pairs not listed are oriented the other way. Its scores are 4, 4, 2, 2, 1, 2, so it has no source and no sink. Let T_7 be the tournament on $\{1, \dots, 7\}$ with the arcs

$$1 \rightarrow 2, 3, 4, 5, 6; \quad 2 \rightarrow 3, 4, 5, 6, 7; \quad 3 \rightarrow 4, 5, 7; \quad 4 \rightarrow 5, 6, 7; \quad 5 \rightarrow 6; \quad 6 \rightarrow 3; \quad 7 \rightarrow 1, 5, 6,$$

with scores 5, 5, 3, 3, 1, 1, 3; it has no source and no sink either.

Lemma 3.1. *T_6 and T_7 are link-irregular.*

Proof. Table 1 lists the score sequence of every vertex-deleted subtournament; each entry is obtained by deleting one vertex and recounting out-degrees. For T_7 the seven sequences are distinct, so the seven links are pairwise non-isomorphic. For T_6 the sequences are distinct except for $T_6 - 4$ and $T_6 - 5$, which both have score sequence $(1, 1, 2, 3, 3)$. In $T_6 - 4$ the unique vertex of score 2 is 6, and it beats vertex 1, which has score 3. In $T_6 - 5$ the unique vertex of score 2 is 3, and it beats only the vertices 4 and 6, both of score 1. An isomorphism would carry the vertex of score 2 to the vertex of score 2 and preserve the scores of its out-neighbours, so $T_6 - 4 \not\cong T_6 - 5$. $\square$

TABLE 1. Score sequences (in increasing order) of the vertex-deleted subtournaments of T_6 and T_7 .

v	score sequence of $T_6 - v$	v	score sequence of $T_7 - v$
1	(1, 1, 2, 2, 4)	1	(1, 1, 2, 3, 3, 5)
2	(1, 2, 2, 2, 3)	2	(1, 1, 3, 3, 3, 4)
3	(0, 2, 2, 3, 3)	3	(0, 1, 3, 3, 4, 4)
4	(1, 1, 2, 3, 3)	4	(1, 1, 2, 3, 4, 4)
5	(1, 1, 2, 3, 3)	5	(1, 2, 2, 2, 4, 4)
6	(1, 1, 1, 3, 4)	6	(0, 2, 2, 3, 4, 4)
		7	(1, 1, 2, 2, 4, 5)

Proof of Theorem 1.1. Put $T_{n+2} = T_n^+$ for $n \geq 6$. Starting from T_6 and T_7 , Lemmas 3.1 and 2.1 show by induction on n that T_n is a link-irregular tournament on n vertices without a source or a sink, for every $n \geq 6$. For $2 \leq n \leq 5$ no link-irregular tournament exists, as Bastien and Khormali observed [2, Observation 7]: the n links are tournaments on $n - 1$ vertices, and up to isomorphism there are only 1, 1, 2, 4 tournaments on 1, 2, 3, 4 vertices, fewer than n in each case. $\square$

Remark 3.2. The construction is explicit: T_n is obtained from T_6 or T_7 by adding the pairs $(s_1, t_1), (s_2, t_2), \dots$ in turn, where s_i beats all earlier vertices except t_i , all earlier vertices beat t_i , and $t_i \rightarrow s_i$. The proof of the step uses the absence of a source and a sink in T , and this is exactly what it hands on to T^+ . Going through the 2^{15} tournaments on six vertices shows that T_6 is one of exactly four link-irregular tournaments on six vertices up to isomorphism, and that none of the four has a source or a sink. On seven vertices there are 139, eight of which have a source or a sink. Since an automorphism of a tournament maps each link onto an isomorphic link, a link-irregular tournament has no automorphism other than the identity.

Remark 3.3. Every link-irregular tournament on $n \geq 6$ vertices whose links have pairwise different score sequences gives such a tournament on $n + 2$ vertices with the same property, because the score sequence of $T^+ - v$ is that of $T - v$ with every entry raised by one and the entries 1 and $n - 1$ added. Starting from T_7 , this gives link-irregular tournaments of every odd order whose links are already told apart by their score sequences.

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